

CSA4302 IP
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1. Online Symposium Portal

1. Identify missing semantic elements

The structure can be improved using:

- **<header> – symposium title**
- **<nav> – navigation**
- **<main> – main content**
- **<section> – registration section**
- **<footer> – footer information**
- **<label> – accessible form labels**

2. Standards-compliant structure

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Symposium Registration</title>

</head>

<body>

<header>

<h1>College Symposium</h1>

</header>

<main>

<section>

<h2>Registration</h2>

```
<form action="/register" method="POST">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name"
required>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email"
required>

  <button type="submit">Submit</button>
</form>
</section>
</main>

<footer>
  <p>© 2026 College Symposium</p>
</footer>

</body>
</html>
```

3. HTTP Request

When Submit is clicked, the browser sends:

POST /register HTTP/1.1

Content-Type: application/x-www-form-urlencoded

name=Sanjai&email=sanjai@example.com

POST sends form data in the request body.

4. HTTP Response Cycle

Browser → HTTP Request → Server

Browser ← HTTP Response ← Server

The server processes the registration and returns a response such as 200 OK. The browser then displays the result.

5. Usability and Accessibility

- Use <label> for inputs.**
- Use required fields.**
- Provide clear error messages.**
- Support keyboard navigation.**
- Use proper color contrast.**
- Make the page mobile-friendly.**

2. Cross-Browser Compatibility Issue

1. Improper nesting and missing tags

Given:

<h1>Welcome

<p>About us

Missing:

</h1>

</p>

Also, <!DOCTYPE html> is missing.

2. Browser Interpretation

Browsers may automatically correct malformed HTML differently, causing:

- Different DOM structures**
- Different spacing**
- Different layouts**
- CSS/JavaScript problems**

3. Missing Semantic Elements

Use:

- <header>**
- <main>**
- <section>**
- <footer>**

instead of using only <div>.

4. Correct HTML

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Company</title>

</head>

<body>

<header>

<h1>Welcome</h1>

</header>

<main>

<section>

<h2>About Us</h2>

<p>About our company.</p>

</section>

</main>

<footer>

<p>© 2026 Company</p>

</footer>

</body>

</html>

5. Cross-Browser Techniques

- Follow W3C/HTML5 standards.**
 - Use valid HTML and CSS.**
 - Use <!DOCTYPE html>.**
 - Test in Chrome, Firefox, Edge and Safari.**
 - Avoid deprecated browser-specific features.**
 - Use responsive design.**
-

3. Form Submission Failure

1. Issues

<button type="button">Login</button>

is incorrect for form submission.

Also, GET should not be used for passwords.

2. Why HTTP Request Is Not Triggered?

type="button" is a normal button and does not submit the form.

Use:

<button type="submit">Login</button>

3. Role of GET

GET sends form data through the URL:

/submit?username=Sanjai&password=12345

Therefore, GET is unsuitable for passwords because sensitive data can appear in browser history, URLs, and logs.

4. Correct Implementation

<form action="/submit" method="POST">

<label for="username">Username:</label>

<input type="text" id="username" name="username" required>

<label for="password">Password:</label>

<input type="password" id="password" name="password" required>

<button type="submit">Login</button>

</form>

5. Secure Data Transmission

- Use HTTPS.**
- Use POST for login credentials.**
- Validate data on the server.**
- Hash passwords securely.**
- Use secure authentication/session management.**

Final Point

HTML5 provides structure and accessibility, while HTTP enables communication between the browser and server. Proper HTML, form methods, and secure HTTP practices ensure reliable and secure web applications.